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Oefeningenreeks 3H nr. 1.6

$$x^2 + y^2 = 2x(3y^2 - x^2)$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$\Leftrightarrow r^2 \cos^2 \theta + r^2 \sin^2 \theta = 2r \cos \theta (3r^2 \sin^2 \theta - r^2 \cos^2 \theta)$$

$$\Leftrightarrow r^2 = 2r \cos \theta \cdot r^2 (3 \sin^2 \theta - \cos^2 \theta)$$

$$\Leftrightarrow r^2 = 2r \cos \theta \cdot r^2 (3 \sin^2 \theta - \cos^2 \theta)$$

$$\Leftrightarrow \frac{1}{r} = 2 \cos \theta (3 \sin^2 \theta - \cos^2 \theta) \vee r = 0$$

$$\Leftrightarrow r = \frac{1}{2 \cos \theta (3 \sin^2 \theta - \cos^2 \theta)} \vee r = 0$$

## References